

Java Exception Handling Lab Report

Program: InputMismatchException

1. Aim

To demonstrate handling of InputMismatchException in Java when the user enters an invalid input instead of a whole number.

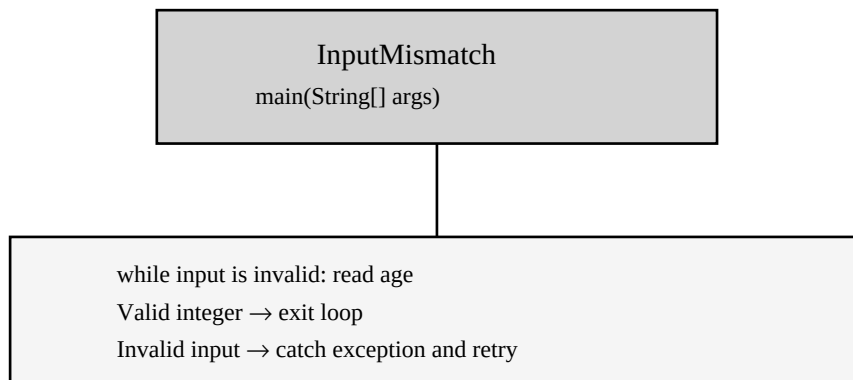
2. Description

InputMismatchException occurs when a Scanner method receives input that does not match the expected data type. In this program, nextInt() expects a whole number. If the user enters invalid data, the catch block displays an error message, and scanner.next() removes the invalid input so the loop can continue. The program repeats until a valid age is entered.

3. Tools Used

- Java Development Kit (JDK)
- Java compiler (javac)
- IDE or command prompt
- Scanner class from java.util package

4. UML / Program Structure



5. Algorithm

1. Start the program.
2. Create a Scanner object to read input.
3. Initialize age and the valid flag.
4. Repeat while valid is false.
5. Prompt the user to enter their age.
6. Try to read an integer using nextInt().
7. If the input is valid, set valid to true.
8. If the input is invalid, catch InputMismatchException, display an error message, and remove the invalid input using next().
9. Display the entered age.

10. Close the Scanner and stop the program.

6. Complete Java Source Code

```
package exceptions;
import java.util.InputMismatchException;
import java.util.Scanner;

public class InputMismatch {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int age = 0;
        boolean valid = false;

        while (!valid) {
            System.out.print("Enter your age : ");
            try {
                age = scanner.nextInt();
                valid = true; // Input was valid, exit the loop
            } catch (InputMismatchException e) {
                System.out.println("Invalid input! Please enter a whole number");
                scanner.next();
            }
        }

        System.out.println("Your age is: " + age);
        scanner.close();
    }
}
```

7. Sample Output

```
Enter your age : abc
Invalid input! Please enter a whole number
Enter your age : 19
Your age is: 19
```

8. Result

The program successfully demonstrated InputMismatchException handling. Invalid input is caught and removed, allowing the user to enter another value until a valid whole number is provided.